

ACTL3142 Week 3 - Multiple Linear Regression

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Relate p covariates linearly and additively to a quantitative response. That is,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon.$$

Estimation of coefficients, (most) assumptions, interpretation of coefficients all stays the same!

Estimating Coefficients (same as SLR)

Given a training set, we need assumptions for the ϵ term to estimate the β params.

Ordinary Least Squares (OLS) (weak assumptions)

- Assume errors have zero mean, constant variance and conditionally uncorrelated.
- Don't need distributional assumption for OLS.

MLE (strong assumptions)

- Assume $\epsilon_i \mid \mathbf{x} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.
- Need a distributional assumption to construct our likelihood function and later construct confidence intervals.

Encoding Categorical Variables

Suppose a *categorical* predictor z can take on q different values. We need an alternative way to represent this in our model.

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Dummy-Encoding

1. Pick one possible value of z and make this the “base case.” It will be reflected in β_0 .
2. With the remaining $q - 1$ possible classes, make new predictors $x_1, x_2, \dots, x_{q-1}$, which have value 1 if z is the respective category, otherwise 0.
3. Estimate $q - 1$ coefficients.

It is important to remember we only estimate $q - 1$ coefficients even though z can take on q unique values. **The resulting estimates are with respect to the base case.**

Comparing Different Models

We need ways to compare models with different predictors, and determine whether including features is actually adding valuable information or if we're just fitting to noise.

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- F -test - test $\mathcal{H}_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$, i.e. if the entire model is actually significant;
- Adjusted R^2 - similar to R^2 , but penalty for adding more predictors (higher is better);
- C_p is an estimate for test error, adding penalty for more predictors (lower is better).
- AIC and BIC are alternates to log-likelihood which adds penalty for more predictors (lower is better).

Forward Stepwise Selection

1. Start with the null model $\mathcal{M}_0$.
2. For $k \in [0, p - 1]$: Consider all $p - k$ models which add one predictor to $\mathcal{M}_k$ and pick the best one via R^2 .
3. Pick the best out of all the models found above via adjusted R^2 , C_p and BIC.

Backward Stepwise selection is the reverse, where we begin with $\mathcal{M}_p$ and remove predictors that cause the least reduction in R^2 .

Best subset selection considers all possible models.

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Why do we expect the 'best' model to be somewhere in between $\mathcal{M}_0$ and $\mathcal{M}_p$?

Shortfalls in Linear Regression

We make a lot of assumptions in linear regression! Below are common violations/shortfalls and how to identify them.

1. Non-linearities between predictors and response.
→ Patterns in residual plots.
2. Correlations in error terms.
→ Correlations in error terms across fitted values.
3. Heteroskedastic errors (as opposed to homoskedastic).
→ Funnel shape in residual plots.
4. Outliers (violation of normal errors)
Very different values in residual plots.
5. High-leverage points
→ Large Cook's distance.
6. Collinearity (violation of additive assumption)
→ High values of VIF.